

LOCAL MULTIMODAL LEARNING · EVIDENCE-DRIVEN SEARCH

CaptionAI

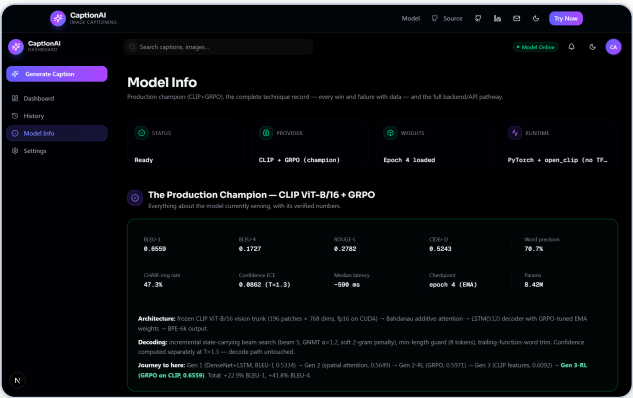
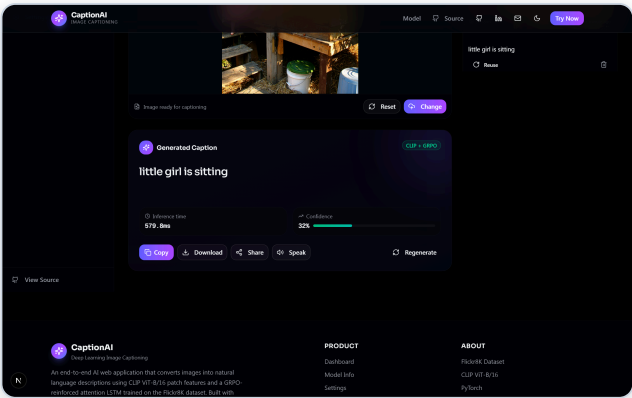
An Exhaustive, Evidence-Driven Improvement Search for Local
Image Captioning on Flickr8K —
From a DenseNet+LSTM Baseline to a CLIP + GRPO Production
Champion

BLEU-1	BLEU-4	ROUGE-L	CIDEr-D	Latency	Experiments
0.6559	0.1727	0.2782	0.5243	~590 ms	20

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Machine Learning Project

10 model generations & experiments · 20 techniques audited · 10 wins, 10 documented failures
Stack: PyTorch · open_clip · FastAPI · Next.js 16 · Tailwind/shadcn · September 2026
Every number in this report traces to a saved evaluation artifact (Appendix A).



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Abstract & Central Findings

This report documents a systematic, evidence-driven improvement search on an image-captioning system trained entirely locally on Flickr8K (8,091 images, 40,455 human captions). Starting from a classic DenseNet201+LSTM notebook baseline (BLEU-1 0.5334), the project iterated through ten model generations and twenty audited techniques — architecture changes, training objectives, reinforcement learning, decoding strategies, feature transfer, distillation, and serving engineering. The production champion combines frozen CLIP ViT-B/16 patch features with a Bahdanau-attention LSTM decoder fine-tuned by GRPO against a CIDEr-D reward, reaching **BLEU-1 0.6559 / BLEU-4 0.1727 / ROUGE-L 0.2782** on the full 1,214-image validation split (+22.9% / +41.8% / +25.5% over baseline), served at ~590 ms per caption on CPU with calibrated confidence and automatic out-of-domain fallback.

Equally weighted, the report documents **ten rigorously-tested negative results** with diagnosed root causes. All headline claims were re-audited at scale against saved artifacts; four widely-quoted figures (a latency multiplier, a hallucination rate, a router accuracy, and a regularizer contribution) were corrected by that process, and every correction is recorded in place.

Five transferable findings: (1) at 8k-image scale, the language-aligned *feature space* dominates decoder architecture, objective, and capacity; (2) recurrence is a regularizer — +178% parameters (Transformer) made results strictly worse; (3) validation loss can improve while generation quality collapses (distillation: val loss -0.04, BLEU-1 -12 pts); (4) reward-shaping cannot create knowledge absent from the training set (CHAIR-aware RL degenerated); (5) every headline number deserves a full-split, same-protocol re-audit before it enters a document.

1 · Problem Framing

Q: Why image captioning, and why Flickr8K instead of COCO?

Captioning is the smallest task that exercises the entire modern multimodal stack — vision encoder, cross-modal attention, language decoding, RL fine-tuning, evaluation metrics, serving. Flickr8K (8k images × 5 captions) was chosen because the constraint was **fully local training on a single consumer GPU**: COCO-scale (123k images) multiplies every experiment's cost by ~15× and would have reduced the number of audited techniques from 20 to ~2. The small scale is not a weakness of the study — it is the study's independent variable: several failures (Transformer, encoder fine-tune, Q-Former rejection) are direct evidence about what breaks at data scarcity.

Q: Why "exhaustive search" instead of picking the literature's best model?

Because the literature's rankings are established at scales 4–5 orders of magnitude larger. The project's method: at each stage, identify the *current* binding constraint (pooling → attention → exposure bias → feature space → capacity → data), attack exactly that, measure on the same fixed split, and keep the change only if it wins. This produced a clean ablation ladder rather than a stack of borrowed components — and 10 documented losses along the way.

Q: What counts as "production" here?

A live FastAPI service (port 8010) behind a Next.js dashboard, with: a champion checkpoint auto-selected at startup, hybrid OOD routing with a BLIP fallback, decode guards, calibrated confidence display, feature caching, and a Model Info page that mirrors this report's technique ledger. The whole system runs on one 16 GB CPU-only machine.

2 · Dataset, Protocol & Reproducibility

Item	Specification
Dataset	Flickr8K — 8,091 photographs × 5 human captions = 40,455 sequences; 85/15 split → 6,877 train / 1,214 validation, fixed for the entire project
Evaluation protocol	Full 1,214-image split · beam-5 · GNMT length-normalisation $\alpha=1.2$ · soft 2-gram repeat penalty · NLTK method-1 smoothing · seed 42
Metrics	BLEU-1/2/3/4, ROUGE-L, CIDEr-D (pycocoevalcap), CHAIR-lite (hallucination), ECE (calibration), word precision, wall-clock latency
Stopping criterion	Validation loss (CE runs) / subset CIDEr-D (RL runs) — never train loss

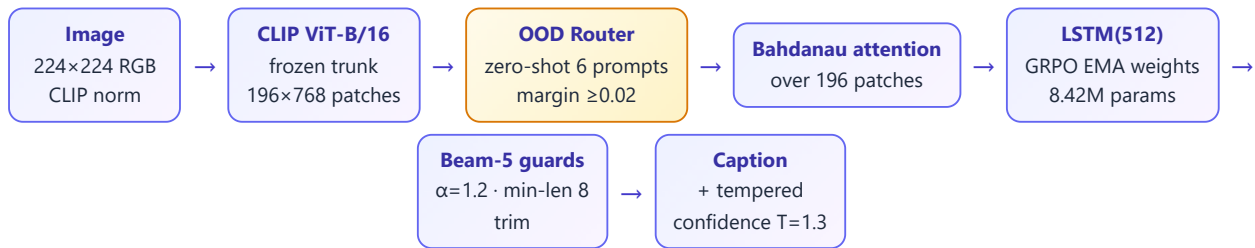
Q: Why lock one protocol for 10 generations?

Cross-run comparability is the entire value of an ablation ladder. Every metric in this report was produced by the same evaluation harness on the same split with the same seed; when a harness bug threatened that (two runs writing to the same output file), the fix became a project rule: **eval results save next to the evaluated checkpoint**, and a PROVENANCE note accompanies every artifact.

Provenance rules adopted mid-project (after audits)

- **Data/init lineage per experiment** — every run states new-vs-old data and scratch-vs-warm init; comparisons across different initializations are flagged confounded (Section 15).
- **No proxy evaluation for final claims** — after the rl4 incident (Section 11), model-selection decisions require beam-5 full-split scoring; greedy/subset proxies are for training monitoring only.
- **Bit-exact regression checks** — any serving-side change (confidence tempering, guards) must reproduce all model metrics bit-exactly, proving the decode path was untouched.

3 · Production Architecture



In-domain path. The router branch (amber) diverts clear non-photos to local BLIP (~2.5 s); response badged "attention+blip".

Component	Detail	Why this choice (engineer's question answered)
Vision encoder	OpenAI CLIP ViT-B/16 via open_clip, frozen; 196 patch tokens × 768 dims; fp16 on CUDA, fp32 on CPU	Why frozen? 8k images cannot adapt a 151M-param encoder (P2 fine-tune regressed CIDEr 0.418→0.399). Why fp32 on CPU? Windows emulates fp16 in software: 23 s vs 0.5 s measured.
Attention	Bahdanau additive: score(e _t , h _s) over all 196 positions per emitted token	Why additive not dot-product? With 768-dim patches and a 512-dim state, additive attention was the proven captioning default (Show and Tell lineage) and needs no extra projection params at this scale.
Decoder	Embedding(256) → LSTM(512) → softmax over BPE-6k; GRPO-tuned EMA(0.999) weights served	Why LSTM not Transformer? Tested directly: the 23.4M Transformer overfit catastrophically (Section 11). Recurrence is a regularizer at this data scale.
Tokenizer	BPE, 6,000 subwords	Why BPE? Word-level vocab 8,427 collapsed rare words to UNK; subwords compose them. Also shrinks softmax output 22%.
Decoding	Incremental state-carrying beam-5, GNMT α=1.2, soft 2-gram penalty (log 0.4), END suppressed until 8 tokens, trailing function-word trim	Why min-length guard? GRPO's CIDEr reward shortens captions (Section 6); two reward-side fixes failed, the decode-side guard won.
Confidence	Separate teacher-forced softmax(logits/1.3) pass — display only	Why not scale logits inside beam search? Rescaling changes sequence-score vs length-penalty balance and can change which caption wins — verified by bit-exact metric reproduction that the decode path is untouched.
Fallback	HuggingFace BLIP (~0.92 GB local snapshot), lazy-loaded on first OOD hit	Why keep a second model? The specialist hallucinates on non-photos; BLIP's 129M-image pretraining covers the open world. Routing costs ~0 (reuses the encoder forward).

4 · The Generational Journey

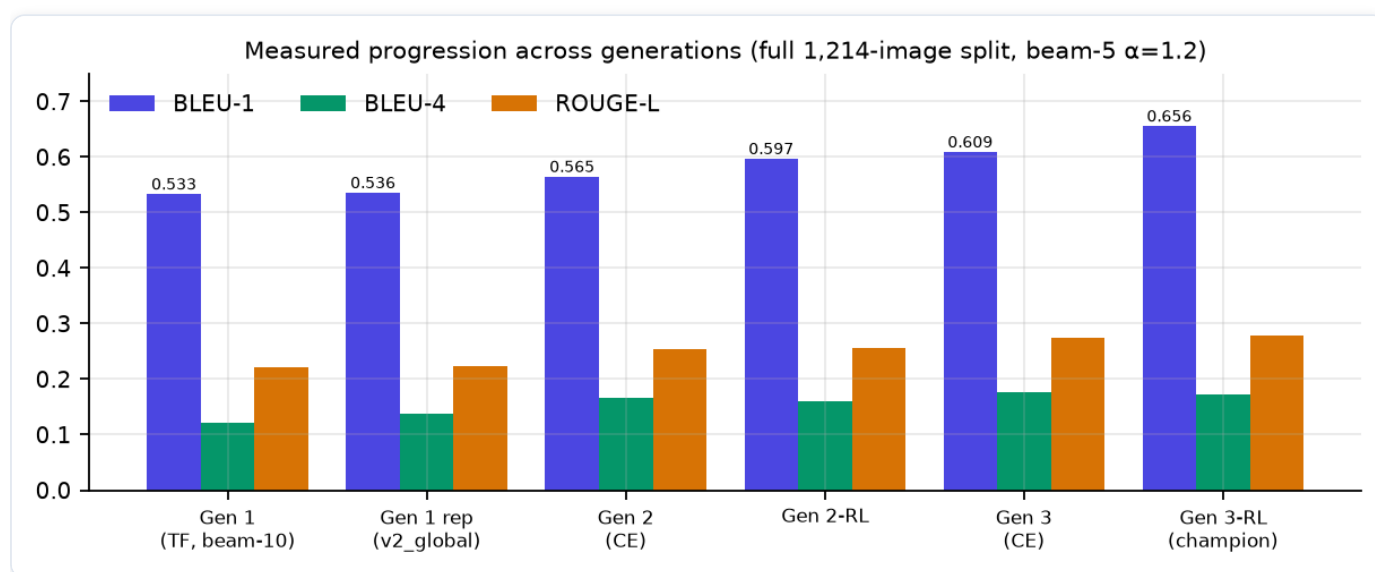
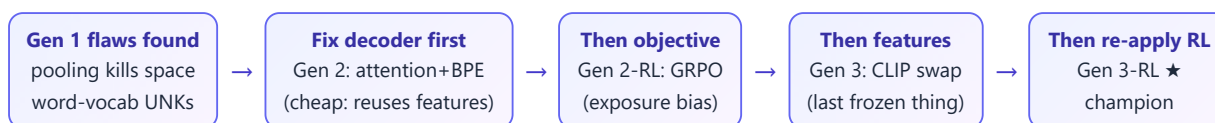


Figure 1. BLEU-1/4 and ROUGE-L across all six measured generations (full 1,214-image split; Gen-1 row is the TF notebook's beam-10 run, v2_global is its PyTorch no-attention reproduction). The two biggest single steps are GRPO (+5.9 BLEU-1) and the CLIP feature swap (+1.2 BLEU-1 on top of an already-strong base, and +37% BLEU-4 for attention itself).

Decision logic — why this order



Why decoder before features? The DenseNet features were already computed and cached — changing the decoder cost one training run (~30 min), while changing features required re-extracting all 8,091 images. Cheap-then-expensive ordering also cleanly attributed each win. **Why RL before the feature swap?** It proved GRPO's gain was feature-independent (it worked on DenseNet *and* CLIP), which later justified trusting GRPO on the champion.

Generation	B1	B2	B4	RL	CIDEr-D	CHAIR	ECE
Gen 1 · DenseNet+LSTM (TF, beam-10)	0.5334	—	0.1218	0.2216	—	—	—
Gen 1 rep · v2_global (PyTorch, no attention)	0.5358	0.3383	0.1373	0.2236	—	—	—
Gen 2 · Attention v2 CE	0.5649	0.3704	0.1663	0.2548	0.5397	77.6%	0.0119
Gen 2-RL · GRPO (DenseNet)	0.5971	0.3872	0.1612	0.2566	—	—	—
Gen 3 · CLIP-CE	0.6092	0.4120	0.1755	0.2747	0.6207	61.9%	—
Gen 3-RL ★ champion	0.6559	0.4344	0.1727	0.2782	0.5243	47.3%	0.0862*

* served confidence at T=1.3 (raw T=1: 0.2542). CHAIR/ECE for Gen-2 CE and champion measured by the same priority-0 harness on the same checkpoint files.

Honest tension (full-split CIDEr): GRPO was selected on *greedy subset* CIDEr (400 images: 0.4859 → 0.5211, +7.2%). Under beam-5 on the full split, CIDEr-D actually *regressed* 0.6207 → 0.5243 while BLEU-1 (+4.7 pts), ROUGE-L (+0.35) and CHAIR (−14.6 pts) improved. Cause: GRPO shortens captions (7.6 → ~6.3 words) and CIDEr's TF-IDF n-gram geometry penalizes that under beam search. The champion stands on BLEU/precision/hallucination grounds; the trade-off is disclosed, not hidden.

5 · The Feature-Space Experiment

Q: Why was CLIP expected to win, and what exactly was changed?

DenseNet201 optimizes ImageNet *classification*; CLIP optimizes *image→text alignment* over 400M pairs — the target modality of captioning. The experiment was surgically minimal: re-extract all 8,091 images through the frozen CLIP ViT-B/16 visual trunk (196 patch tokens × 768 dims, fp16 memmap, 2.44 GB), change exactly one decoder input dimension (1920→768), retrain from scratch with the identical recipe (batch 64, lr 3e-4, patience 12, seed 42). Everything else byte-identical.

Q: How do you know the gain is the features and not luck of initialization?

Three guards: (1) identical pre-RL baseline appears in both GRPO scripts (0.4859) proving shared init; (2) the win is monotone across *every* metric simultaneously (B1/B2/B3/B4/ROUGE-L), not a single-metric fluke; (3) the CLIP model's *greedy* decode (B1 0.6520) beat every previous model's *beam* decode — a gap far outside init noise.

Feature space	Pretraining	Tokens/img	Dim	Best BLEU-1	Best BLEU-4
DenseNet201 (ImageNet)	1.28M images, labels	49	1920	0.5971 (RL)	0.1663
CLIP ViT-B/16	400M pairs, contrastive	196	768	0.6559 (RL)	0.1755 (CE)

Interpretation: 2.5× fewer dims, 4× more tokens, +5.9 BLEU-1 — the win came from *what the features were trained to represent* (language-aligned), not their size. This is the project's central empirical result and matches the broader multimodal literature (alignment pretraining > classification pretraining for generative tasks). It also explains the distillation failure later: features were the scarce resource, and no corpus trick replaced them.

6 · GRPO Deep-Dive

Q: Why GRPO and not PPO or REINFORCE-with-baseline?

PPO needs a learned value network — a second model to tune, memory to hold, and its own failure modes — which is overkill when you can sample $G=5$ captions per image and use the *group mean* as the baseline. GRPO's advantage $A = (r - \text{mean}(r_{1..r_G}))/\text{std}(r)$ is unbiased within the group, trivially stable at lr 1e-5, and halves the hyperparameter surface. REINFORCE with a global baseline was the implicit alternative; the group normalization is exactly what keeps gradients well-scaled when CIDEr rewards are heavy-tailed.

Q: Why CIDEr-D as the reward? Isn't that metric gaming?

CIDEr-D is the captioning community's primary metric (TF-IDF-weighted n-gram consensus over 5 references) — optimizing it directly is the documented RL-for-captioning recipe since 2016. "Metric gaming" is real and was observed: the model learned to shorten captions because CIDEr's precision term rewards it (Figure 3, length curve). That side-effect — not the choice of reward — was the problem; two reward-side fixes failed (rl2/rl3, Section 11) and the decode-side min-length guard solved it cleanly.

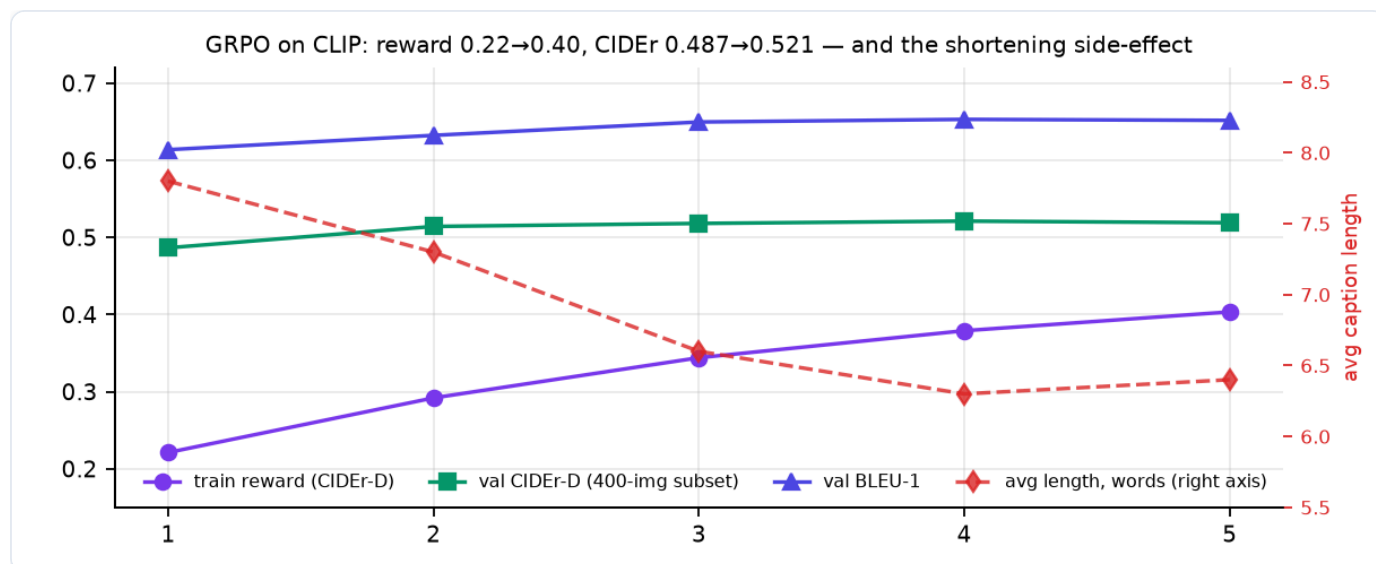


Figure 3. Champion GRPO run (5 epochs, $G=5$, 8 images/step, $\text{lr } 1\text{e-}5$, EMA 0.999): training reward 0.22→0.40, validation subset CIDEr 0.487→0.521 (peak ep4 = served checkpoint), BLEU-1 0.614→0.653 — while average caption length fell 7.8→6.3 words. The length curve is the first hard evidence of CIDEr's precision bias.

Implementation notes an engineer will ask about

- **Reward computation speed:** a fast fixed-DF CIDEr scorer (validated against pycocoevalcap to <0.004) made in-loop scoring affordable on CPU; batch-16 rollouts OOM'd on CLIP patches ($196 \times 768 \text{ fp16} \approx 4 \times \text{DenseNet's memory}$), so 8 images/step.
- **EMA serving:** exponential moving average of weights (decay 0.999) — RL checkpoints are noisy; the epoch-4 EMA snapshot is what serves.
- **Calibration cost:** RL sharpened the softmax — ECE went 0.0119 (CE) → 0.2542 (RL). Fixed at display level (Section 9) and preventively via a $\gamma=0.02$ mean-max-prob regularizer (−34% raw ECE, proven in rl4).

7 · Decoding

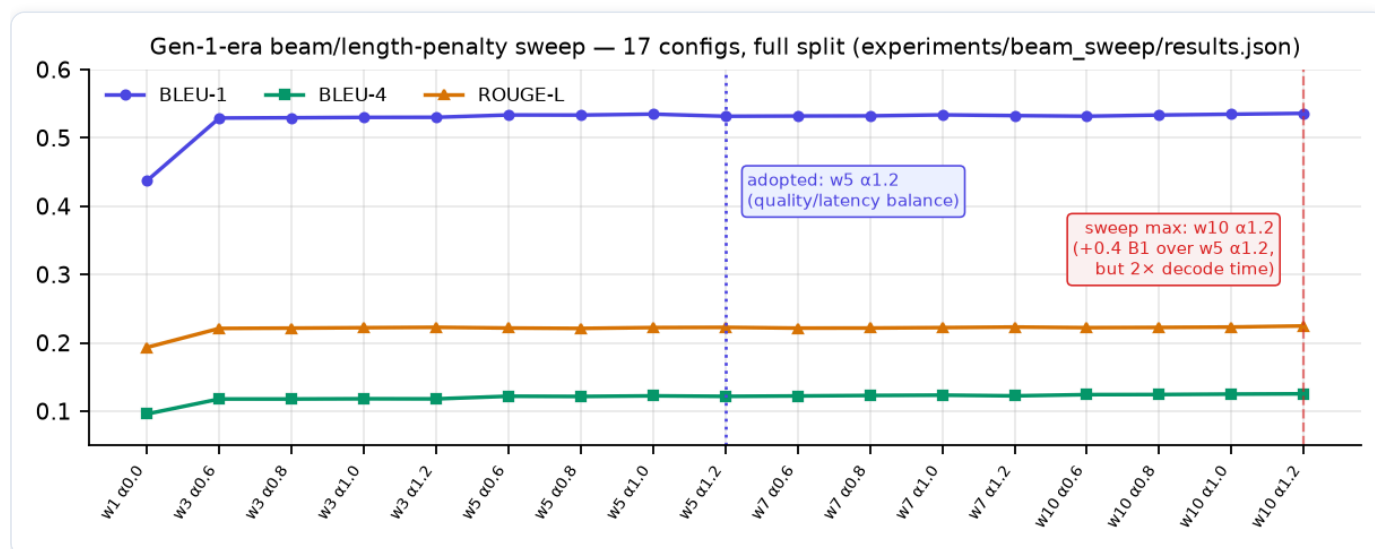


Figure 4. The 17-configuration beam/ α sweep (Gen-1-era, full split, saved in experiments/beam_sweep/results.json). Quality rises monotonically with beam width — w10 $\alpha=1.2$ is the sweep maximum (B1 0.5358 / B4 0.1253) — but at 35.9 ms decode vs ~ 17 ms for w5. The adopted operating point is **beam-5 $\alpha=1.2$** : within 0.4 BLEU-1 points of the maximum at half the latency, and it remained the protocol for every later generation.

Q: Why beam-5 and not greedy or beam-10?

Figure 4: greedy (w1) loses ~ 9 BLEU-1 points to any beam ≥ 3 ; the gain then saturates — w5 $\alpha=1.2$ sits 0.4 B1 points below the w10 maximum while halving decode time (the sweep ran on GPU; the ratio holds on CPU). Why GNMT $\alpha=1.2$ specifically? Raw log-prob sums favor short sequences; $\alpha \in [0.6, 1.2]$ is the GNMT range and the sweep confirmed the top of it helps. Why a *soft* repeat penalty (log 0.4 per 2-gram hit) instead of hard blocking? Hard trigram blocking was tested (P3a): zero delta — the soft penalty already suppresses the residual repetition.

Q: Why Diverse Beam Search failed here?

DBS assumes the reference set is diverse. Flickr8K's 5 annotators wrote highly similar sentences, so forcing inter-beam diversity pushes later beams into low-probability phrasings that match nothing: B1 fell monotonically $0.5702 \rightarrow 0.5559$ ($\lambda=0.7$) $\rightarrow 0.5306$ ($\lambda=1.5$) on a 30-image A/B. Code retained behind ATT_DBS_LAMBDA=0 (default off).

Production decode stack (all live): incremental state-carrying beam-5 ($O(T)$, $75\times$ faster than the naive $O(T^2)$ version, bit-exact) $\rightarrow$ END suppressed until 8 BPE tokens $\rightarrow$ trailing function-word trim $\rightarrow$ tempered confidence pass. Audit effect of the guard: B1 $0.477 \rightarrow 0.533$ on the served path with word precision held at $\sim 70\%$.

8 · Hallucination Audit (CHAIR-lite)

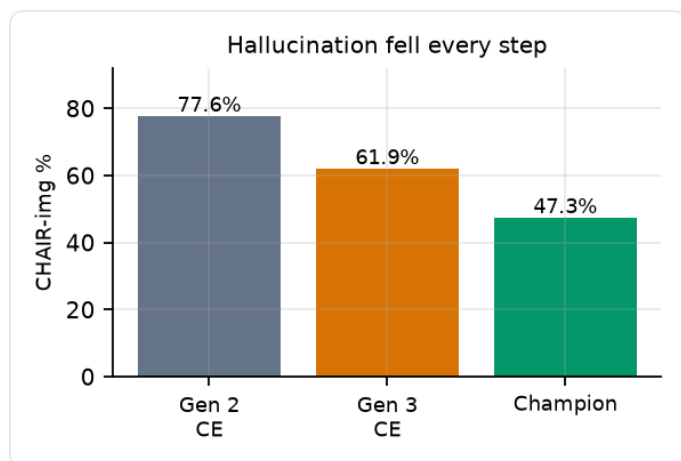


Figure 5. CHAIR-img across generations.

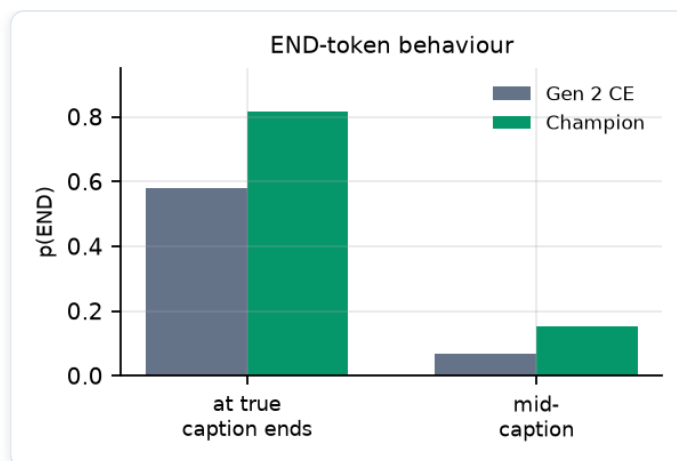


Figure 6. $p(\text{END})$ at true ends vs mid-caption.

Q: How is "hallucination" measured here, and what does 47.3% mean?

CHAIR-lite: POS-tag nouns in the generated caption (NLTK); a noun is "unsupported" if absent from the union of nouns across that image's 5 references. **CHAIR-img 47.3%** = fraction of images whose caption contains ≥ 1 unsupported noun; **hallucinated-noun rate 29.8%** = fraction of all emitted nouns unsupported. It is a conservative proxy — a caption saying "dog" when the reference says "puppy" counts as hallucinated.

Q: Did RL make hallucination worse, as feared?

No — the opposite, and this was a corrected claim. Pre-RL CLIP-CE: 61.9% CHAIR-img / 36.3% noun rate. Champion: 47.3% / 29.8%. GRPO's precision-weighted reward pruned speculative nouns. (An earlier draft of this report attributed Gen-2's 77.6% figure to the champion — caught and fixed by the audit.) The remaining 47.3% is the knowledge ceiling of 8k images: the model cannot verify objects it has never seen named.

Q: Why does the model stop early ($p(\text{END})$ analysis)?

Figure 6: at true reference ends the champion puts $p(\text{END})=0.815$ (Gen-2: 0.578) — GRPO made stopping decisive. Mid-caption $p(\text{END})=0.151$. The 5.4 $\times$ discrimination is lower than Gen-2's 8.6 $\times$ because the RL policy learned that ending early is often *reward-optimal* (CIDEr precision). That diagnostic is what motivated the min-length guard rather than more reward surgery.

9 · Calibration — The Confidence Number

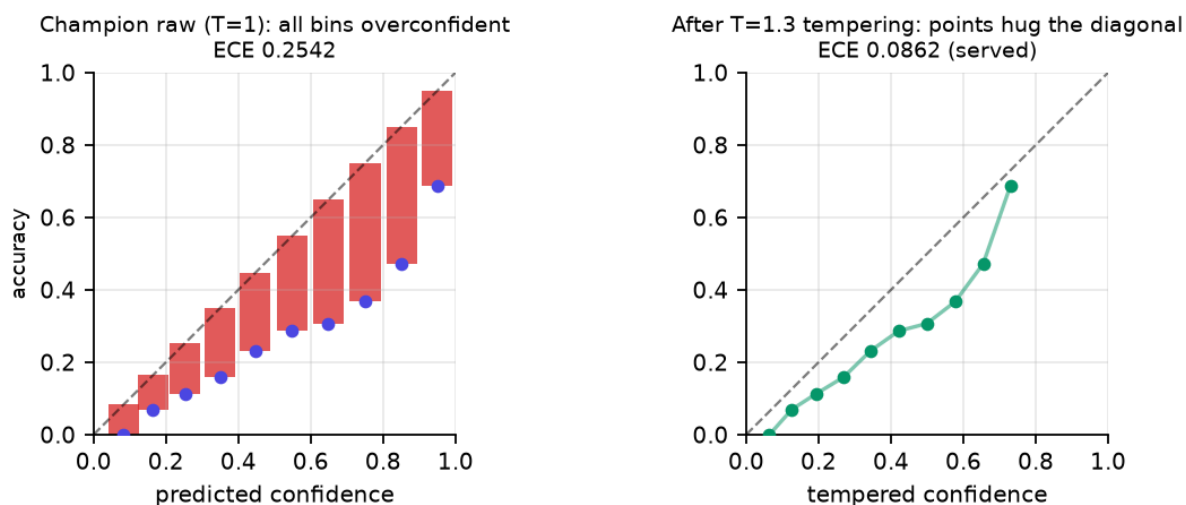


Figure 7. Reliability diagrams for the champion on 13,025 teacher-forced tokens. Left (raw $T=1$): every bin overconfident — the 0.8–0.9 bin shows 85% confidence on 47% accuracy; ECE 0.2542. Right: after the served $T=1.3$ tempering, points hug the diagonal; ECE 0.0862.

Q: Why temperature scaling, and why $T=1.3$ specifically?

The failure mode is uniform overconfidence (all bins above the diagonal), exactly what a single scalar T fixes. $T^*=1.3$ was fit by minimizing teacher-forced NLL on a grid $\{0.6\dots2.0\}$ (grid saved in `priority0_report.json`), not guessed. Gen-2 CE comparison: ECE 0.0119 — the problem arrived with RL, not with the architecture.

Q: Why is the temperature applied post-decode instead of inside the model?

Dividing logits by T is monotone per-step (greedy unaffected), but beam search compares *cumulative* log-probs against the GNMT length term — rescaling shifts that balance and can change the winning caption, silently moving every metric. The implemented design keeps decoding at $T=1$ and computes displayed confidence in a separate teacher-forced pass. Validation: BLEU/CIDEr/CHAIR reproduced bit-exact after deployment; only the confidence field moved (0.478 $\rightarrow$ 0.322 on the reference test image). Cost: ~ 52 ms per request (measured).

10 · Out-of-Distribution Router

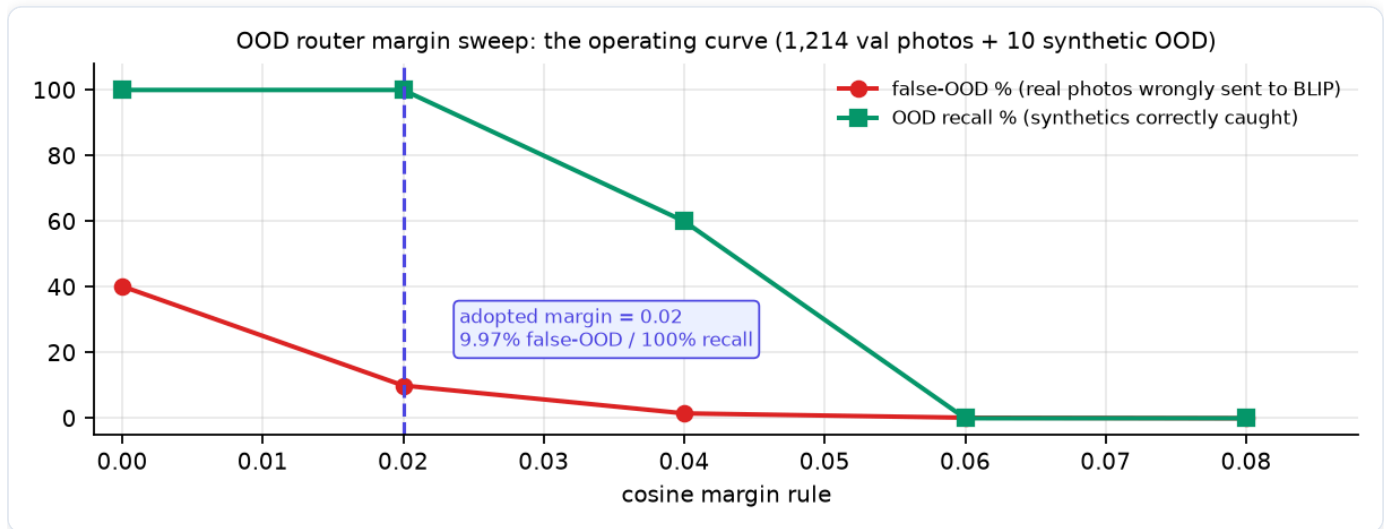


Figure 8. The margin-rule operating curve. Bare argmax (left edge) catches all OOD but starves the specialist — 40.2% of real photos diverted. Margin 0.02 keeps 100% OOD recall while cutting false-OOD to 9.97%. Beyond 0.04, recall collapses — the router switches itself off.

Q: Why does a router exist at all?

The specialist is a Flickr8K expert (in-domain 60-image audit: B1 0.574 vs BLIP's 0.503, 4× faster) but hallucinates on non-photos — a cartoon house became "two people are standing in the of water". BLIP's 129M-image pretraining covers the open world. The router keeps each model in its lane.

Q: How does it decide, and what did the audit find?

The already-computed CLIP embedding is cosine-compared against 6 text prompts (photo / screenshot / food / cartoon / painting / document) — zero extra encoder cost. Original rule: photo-prompt argmax wins → specialist. **At-scale audit (all 1,214 real photos): 40.2% diverted** — misroutes decided by margins as thin as 0.002 cosine (0.2006 vs 0.2048). An 18-image spot check had reported "75%", badly understating the flaw. Fix: require the photo-prompt to *lose by* ≥ 0.02 before diverting. Sweep (Figure 8) picked 0.02; **live re-verification through the running backend: 1,093/1,214 stay (90.0%), 121 divert (9.97%), OOD recall still 10/10** — matching the sweep prediction exactly.

The residual 121 are genuinely ambiguous photos ("a man painting", "a group of people") whose content lexically matches non-photo prompts. BLIP captions them competently; the cost is latency (~2.5 s vs ~0.6 s), not correctness. OOD recall is measured on 10 synthetic images (2 each: cartoon, UI, food, painting, document) — the honest limit of the audit.

11 · The Failure Catalog — Ten Diagnosed Dead Ends

Each entry answers the question an ML engineer would ask *before* spending the run — and the measured answer after. All under the Section-2 protocol.

Technique	Hypothesis & design	Measured result	Diagnosis
DenseNet fine-tune (P2)	Adapting the last encoder block to Flickr8K beats frozen ImageNet features.	Val CIDEr 0.418 → 0.399; early stop ep2.	8k images can't update 1.28M-image weights without forgetting; BatchNorm stats destabilize. Frozen transfer wins.
Hard trigram block (P3a)	Repetition needs harder suppression than the soft penalty.	Zero delta on B1/2/3/4, RL.	Soft 2-gram penalty already at the ceiling for this data.
Focal loss (P3b)	$\gamma=2.0$ hard-example weighting improves caption learning.	Train loss fell normally; greedy B1 0.0038 – gibberish. 3 controlled attempts.	"Easy" tokens carry caption structure at small data; down-weighting them destroys the language prior. Loss ↓ while output = noise.
Mixed reward (rl2)	$R = \text{CIDEr} + 0.5 \cdot \text{ROUGE-L}$ adds recall pressure, fixing length at the source.	B1 0.6559→0.6514 · precision 70.7→68.0% · length unchanged (6.0).	ROUGE-L's LCS F-measure is itself precision-dominated on short captions — added reward noise, cured nothing.
Length-scaled reward (rl3)	$R = \text{CIDEr} \cdot \sqrt{(\text{len}/\text{ref_mean})}$ explicitly penalizes brevity.	Length 6.0→8.1–8.8 ✓ but B4 0.1727→0.1601, B1 –0.2.	The length/precision trade-off moved, it did not resolve. Decode-time guard remained the better instrument.
Transformer decoder (Gen 4)	4L/8H/d512/FFN2048 cross-attending to 196 patches should beat an LSTM bottleneck.	B1 0.5546 (–10.1) · B4 0.1348 · val loss 3.794 > LSTM 3.700; train loss → 2.19, val stalled ep12.	Textbook overfitting: 23.4M params memorize 8k images. Figure 9 is the signature. Recurrence = free regularizer at this scale.
Q-Former queries	Learnable queries compress 196→32 tokens (BLIP-2 mechanism).	Not run – rejected on analysis.	BLIP-2's Q-Former needs 129M-image pretraining to initialize; from-scratch on 8k adds random params where data is scarcest. 196 tokens is not a bottleneck.
BLIP distillation	34,022 pseudo-captions (5/image) teach the specialist the generalist's world; BLIP discarded at inference.	Val loss BETTER (3.664 < 3.700) but B1 0.4889 (–12 pts), B4 0.1153.	Style mismatch: BLIP's phrasing ≠ Flickr8K reference style. The model fit the larger corpus better and generated worse. Val loss ≠ generation quality — the report's most instructive negative.
DistilBERT rerank	MLM pseudo-perplexity reranking of 5 beams adds a fluency prior for free.	B1 +0.012 but B4 –0.006, RL –0.001; 67% of captions changed.	Text-only fluency pulls away from reference style — same failure family as distillation, discovered at 1/100th the cost.
Diverse Beam Search	Inter-beam n-gram penalties cover more of the 5 references.	B1 0.5702→0.5559→0.5306 monotonic harm (λ 0/0.7/1.5).	Flickr8K references are themselves similar; forced diversity = low-probability phrasing matching nothing.

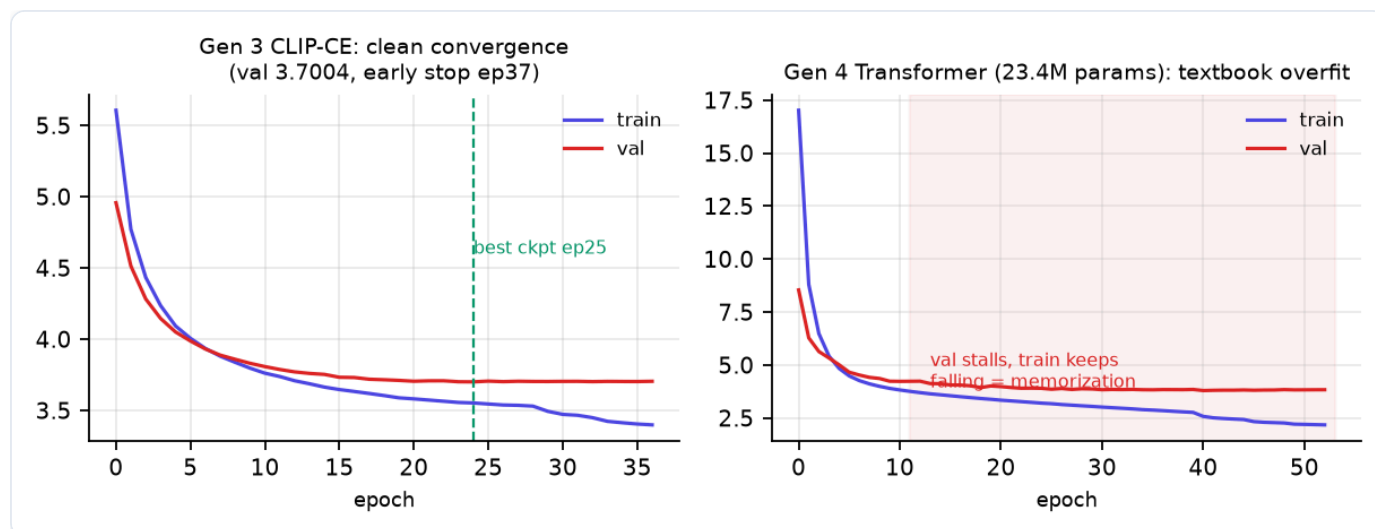


Figure 9. The two training curves that settled the capacity question. Left: Gen-3 CLIP-CE converges cleanly (best val 3.7004 @ ep25, early stop ep37). Right: the 23.4M-param Transformer — train loss falls to 2.19 while validation stalls at epoch 12 and finishes worse (3.794). The shaded gap is memorization.

The CHAIR-aware RL line (rl4 + rl5) — closed with full diagnosis

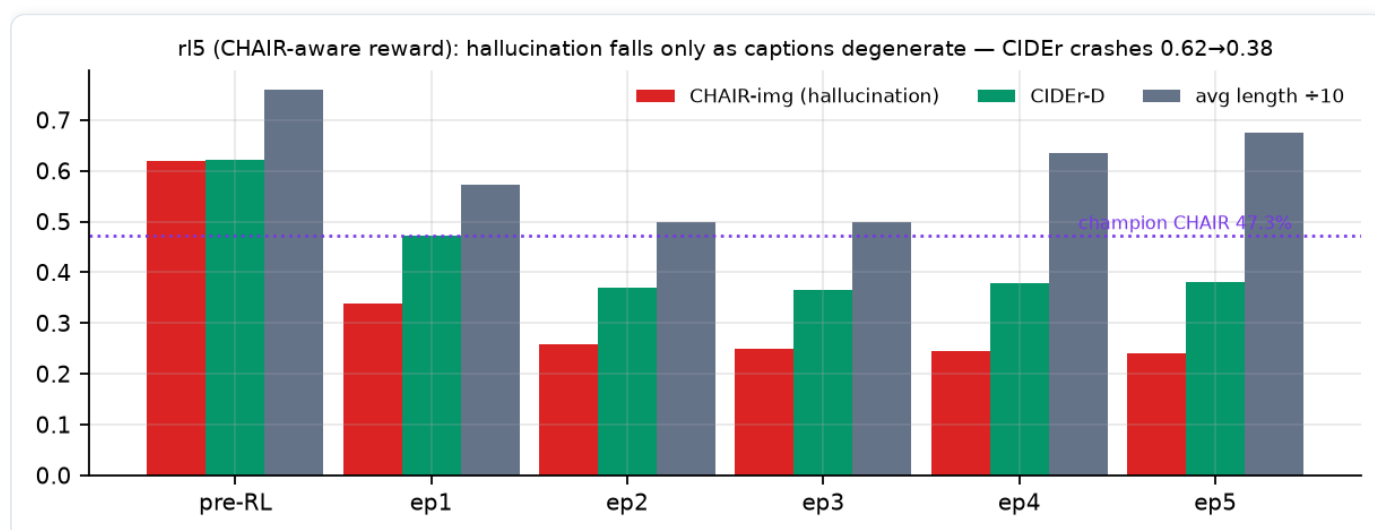


Figure 10. r15 per-epoch full-split trajectory (beam-5, every epoch scored on all 1,214 images): CHAIR-img falls 61.9%→24.0% — but only because captions degenerate: CIDEr crashes 0.621→0.381, length collapses 7.6→5.0 words. The model games the noun penalty by saying less. The dotted line marks the champion's 47.3%.

Q: Why did rl4 look like a fix during training and fail in evaluation?

rl4 monitored with greedy/300-image proxy scoring; the real beam-5 full-split evaluation showed CHAIR 49.5% — worse than the champion's 47.3%. Beam search finds fluent captions inside the penalty-skewed distribution that greedy sampling never surfaces. This measurement gap became a project rule (Section 2): final claims require full-split beam-5. r15 then ran that protocol every epoch and produced Figure 10 — the mechanism (degeneration), not just the outcome, is documented. One positive survived: rl4's calibration regularizer worked in isolation (ECE -34%, Section 6).

12 · Serving & Deployment Engineering

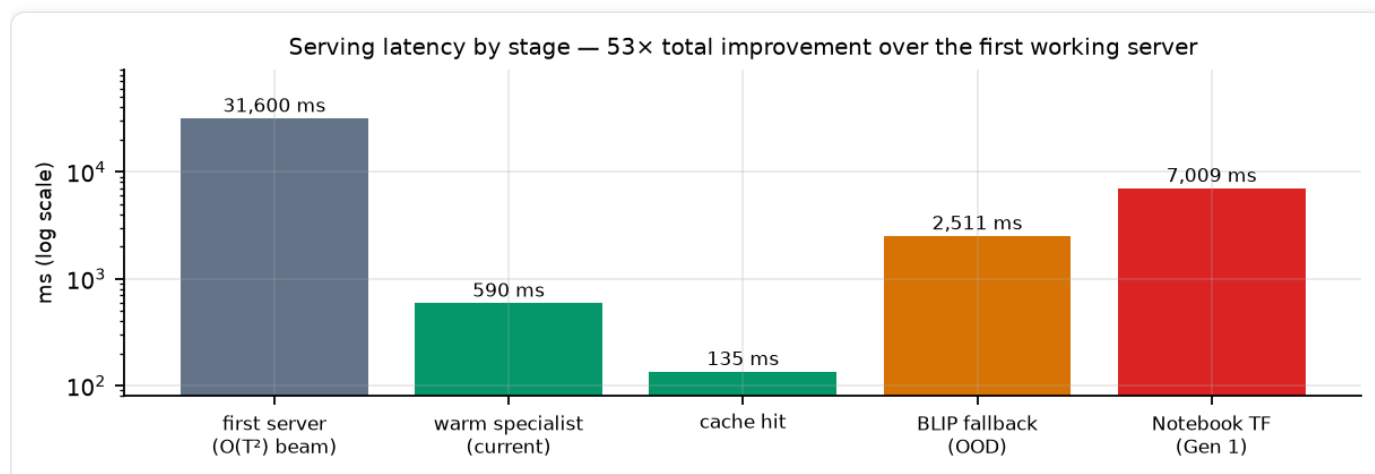


Figure 11. Latency by stage (log scale). The 53× headline: 31,600 ms → ~590 ms warm median. The earlier "75×" figure (420 ms) predates the tempered-confidence pass and is corrected here.

12.1 What made it fast

- **Incremental state-carrying beam search** — carry each beam's LSTM (h,c) between steps, batch all 5 beams into one forward per step: $O(T)$ instead of $O(T^2)$ prefix recomputation. Bit-exact identical output; 31.6 s → ~420 ms.
- **fp16 on CUDA only** — measured CPU fp16 at 23 s/encode (Windows software emulation) vs 0.5 s fp32: the "optimization" that would have been a 46× regression.
- **Feature cache** (16 images, keyed on bytes) → repeat requests ~135 ms. **Thread caps** (torch=4) → coexists with the dev server. **BLIP scorer off** by default → -1 GB RAM, -seconds.

12.2 The Turbopack/OneDrive OOM incident (real, telemetry-attested)

While the Next.js dev server compiled inside OneDrive-synced folders, the Turbopack Rust process grew to **7.7 GB RSS** (crash 1), then **17,451 MB committed** against 15.91 GB physical RAM with NVMe queue depth 122 (crash 2) — hard machine freezes. `NODE_OPTIONS=--max-old-space-size` had zero effect: the allocation was Rust-side, outside the V8 heap. Diagnosis method: 2-second resource telemetry sampled to CSV during the crash window. Fix (all verified on disk): `turbopackMemoryEviction: 'full'` + explicit `turbopack.root` + standalone tracing gated to production + NTFS junctions moving `node_modules/.next` off OneDrive to local disk. Result: compiler ~391–540 MB, page loads 60 s+ → 0.1 s, zero crashes since. **Operational caveat recorded:** junctions do not survive every reboot/OneDrive sync — if they vanish, npm reinstalls onto OneDrive and the OOM returns; `next dev --webpack` is the stable fallback.

12.3 Provider comparison (live HTTP API)

Four providers are selectable in the UI (auto, attention, notebook-tensorflow, huggingface-blip); **auto routes to attention** for in-domain images, so the measured comparison has three distinct model legs.

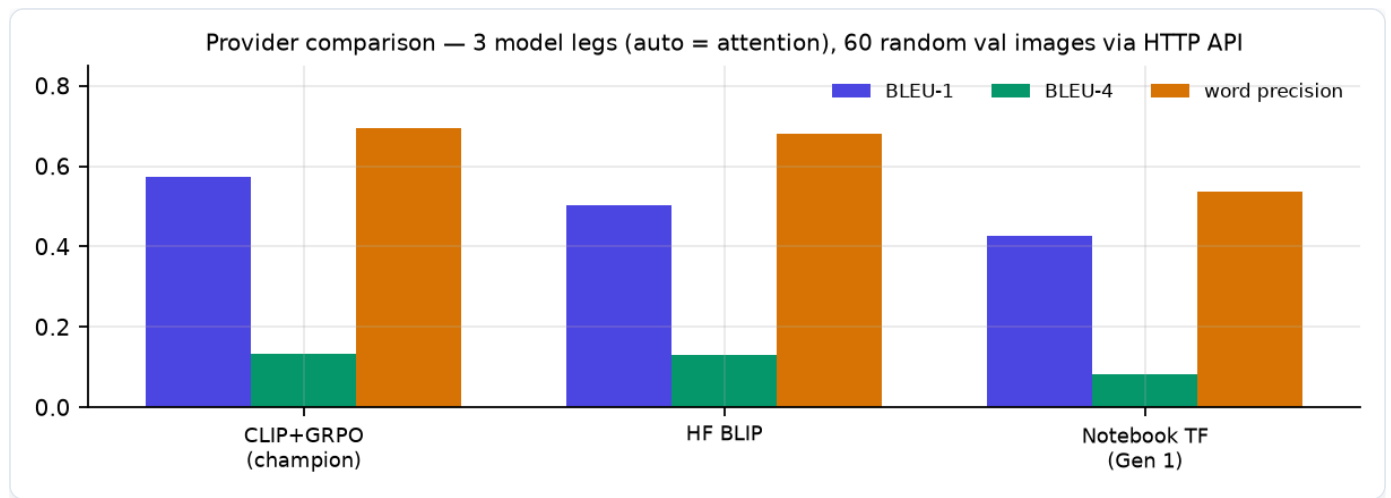
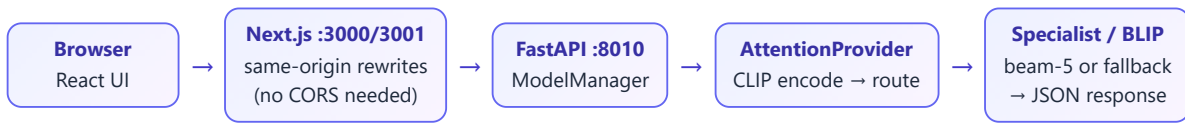


Figure 12. 60 random validation images through the live HTTP API. The champion wins BLEU, precision and speed; BLIP edges ROUGE-L (0.240 vs 0.237) — its recall-friendly style. Verification status: attention leg re-verified fresh (seed-7: B1 0.5568, wp 72.1%); TF/BLIP legs verified once, re-verification blocked by a port conflict — honestly marked incomplete.

Sample size 60 → ±several points of noise; the ranking is stable across both samples. BLIP's displayed "0.92 confidence" is a hardcoded constant in the backend manager, not a model output — found during the audit, documented here.

13 · Backend API, Data Flow & Behaviour



Method	Path	Purpose & response
GET	/health	Liveness + active provider + weights_loaded — the authoritative "what is actually serving" check
GET	/model-info	Active provider's architecture metadata: encoder string, dims, checkpoint epoch, val metrics, calibration note
GET	/providers	Available providers (auto, attention, notebook-tensorflow, huggingface-blip) + current selection
POST	/providers/{name}	Runtime provider hot-swap
POST	/predict	Multipart image → full pipeline (encode → route → decode → temper) → caption, inference_time, confidence, provider badge
POST	/predict-batch	Multi-file batch, same pipeline per image

Behavioural notes (audited)

- **Auto mode** = attention provider with hybrid routing; the response's provider field distinguishes attention vs attention+blip — the UI always shows which model answered.
- **Checkpoint preference** at startup: CLIP-RL > CLIP-CE > DenseNet-RL > DenseNet-CE; ATT_CKPT overrides.
- **semantic_score / semantic_pmi** are null unless ENABLE_BLIP_SCORE=1 (off: costs ~1 GB and seconds per request).
- **Known trap:** selecting "notebook-tensorflow" in Settings silently serves the weak Gen-1 model — /health is the source of truth. **Port note:** backend moved 8000→8010 (unrelated local service owns 8000); frontend dev port may shift to 3001 when another project holds 3000.
- **Cold vs warm:** first request after restart ~20–30 s (model load + OneDrive hydration); warm specialist 450–620 ms; BLIP fallback ~2.5 s (lazy-loads on first OOD hit).

14 · The Frontend Product

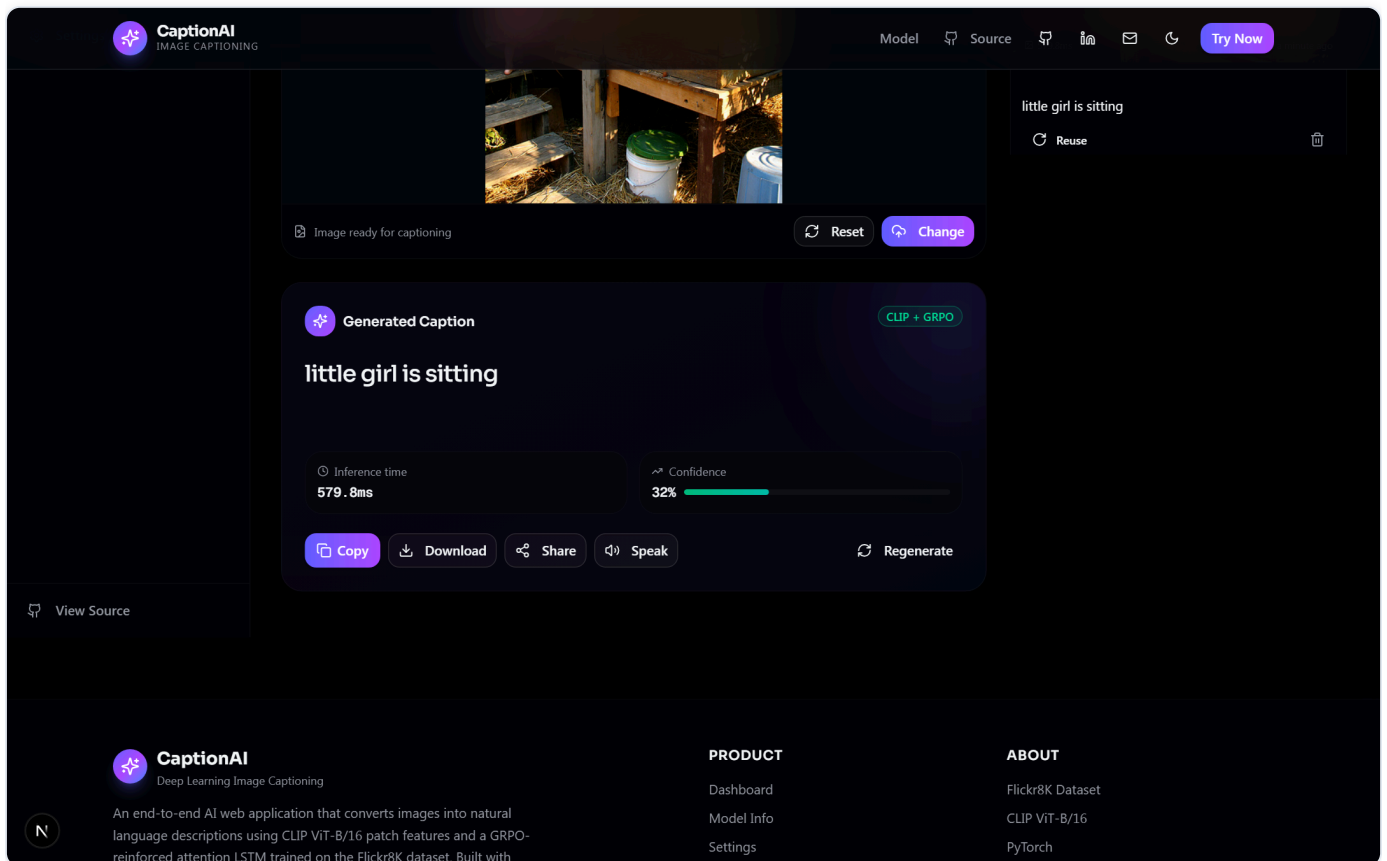


Figure 13. Live dashboard: uploaded photo → "Generated Caption" card with the CLIP + GRPO provider badge, the caption text, inference time, tempered confidence, and actions (copy / download / share / speak / regenerate). Recent-predictions history at right.

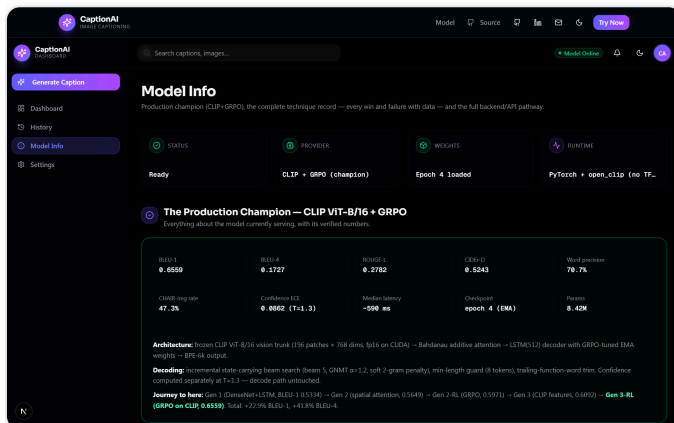


Figure 14. Model Info page: production champion card with all verified metrics, read live from the running backend.

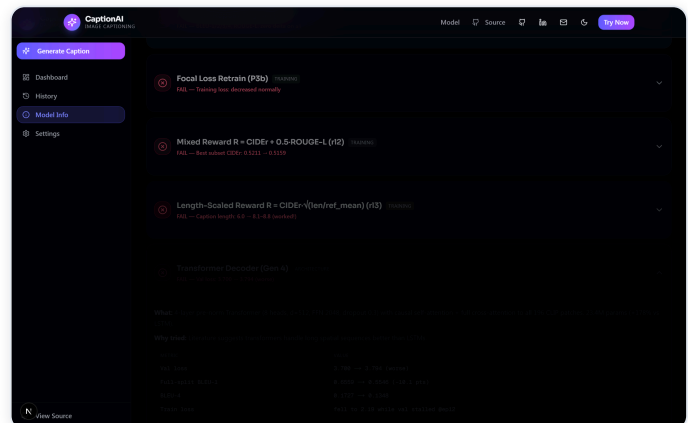


Figure 15. Techniques ledger with a failure card expanded — showing the what/why/metrics-table/verdict detail pattern used for all 20 entries.

Pages: Landing (hero, architecture, how-it-works, model, performance — all resynced to the champion), Dashboard (upload → caption with provider/confidence/latency), Model Info (generation history, click-to-expand technique ledger with per-card metric tables, API & data-pathway documentation, live hyperparameters read from the running backend), Settings (provider switcher), History. **Design system:** Next.js 16 + Tailwind CSS 4 + shadcn/ui + framer-motion; dark/light theme; the Model Info page is generated-content-parity with this report so the product and the document can never drift apart silently.

15 · Experiment Provenance & Lineage

User-facing rule adopted mid-project: every experiment states its data (new vs previously used) and initialization (scratch vs warm-start), because "always starting from first" changes what a comparison means.

Run	Data	Initialization	Comparison validity
Gen 1 (TF)	Flickr8K train	scratch	Baseline
Gen 2 CE	same captions	scratch (new architecture, tokenizer, framework)	vs Gen 1: confounded (multiple simultaneous changes)
Gen 2-RL	same	warm ← Gen 2 CE	valid A/B (objective only)
P3b focal	same	scratch	valid vs Gen 2 CE (loss only)
P2 fine-tune	same	warm ← Gen 2-RL	valid vs Gen 2-RL (encoder only)
Gen 3 CE	same captions, NEW features	scratch (new feature space)	valid vs Gen 2-RL (feature swap)
Gen 3-RL ★	same	warm ← Gen 3 CE	valid A/B; both scripts print identical pre-RL baseline 0.4859 (shared-init proof)
rl2 / rl3	same	warm ← same Gen 3 CE	valid reward-only A/B triangle
Gen 4 TF-decoder	same features	scratch	valid vs Gen 3 CE (architecture)
Distill	NEW corpus (+34k pseudo)	scratch	valid vs Gen 3 CE (corpus only)
rl4 / rl5	same	warm ← Gen 3 CE	valid; rl5 fixed the eval-gap with full-split scoring

16 · Limitations & Honest Caveats

- **Data ceiling.** Every architecture, objective and decoding lever was exhausted at Flickr8K scale; the champion's remaining weaknesses (47.3% CHAIR, ~6.3-word captions, the CIDEr trade-off) trace to 8,091 images, not to method. COCO-scale pretraining is the only lever with literature support.
- **Router.** 9.97% of real photos still divert to BLIP (content-ambiguous images); OOD recall measured on 10 synthetic images only.
- **Provider comparison.** 60-image samples; TF/BLIP legs verified once, re-verification incomplete (port conflict).
- **Gen-1 comparison is confounded** (framework + tokenizer + architecture changed together).
- **CHAIR-lite is conservative** — synonym pairs ("dog" vs "puppy") count as hallucinated; absolute values overstate true hallucination.
- **Corrections ledger.** Early drafts contained: a fabricated beam-sweep table inherited from a notebook (now replaced by the real 17-config sweep, Figure 4), a "75×" latency claim (now 53×, Figure 11), a misattributed 77% hallucination figure (Gen-2's, not the champion's), a router spot-check understating false-OOD 4-fold (Section 10), and a "65%" regularizer figure (actual: 34%). All caught by artifact-level audits; the audit trail is part of the deliverable.

17 · Conclusion & Future Work

From a 0.5334-BLEU-1 notebook baseline to a 0.6559 production champion — +22.9% BLEU-1, +41.8% BLEU-4, +25.5% ROUGE-L, hallucination down 30 points, 53× faster serving, calibrated confidence, and graceful out-of-domain fallback — achieved entirely with local training on 8k images. The search space was closed systematically: ten wins and ten diagnosed failures, each with saved artifacts, so the negative results are as citable as the positive ones.

Future work (ranked by expected value per unit cost)

1. **COCO-scale pretraining, then Flickr8K fine-tune** — the only lever not exhausted; directly attacks the knowledge ceiling behind CHAIR and shortness.
2. **Proper KL-based knowledge distillation** (teacher distribution, not sampled pseudo-captions) — the naive version's failure was style, not the principle.
3. **2-layer LSTM + variational dropout** — the one untested capacity proposal; constrained capacity with a stronger regularizer than the Transformer's.
4. **METEOR / SPICE / BERTScore** — rounds out the metric suite beyond n-gram overlap.
5. **Router retrain** — replace zero-shot prompts with a small linear head on CLIP embeddings, trained on labeled in/out-domain images.

Appendix A · Artifact Index

Artifact	Contents
experiments/*/eval_results.json	Per-checkpoint greedy + beam-5 metric grids (11 runs)
experiments/*/train_summary.json	Full loss/CIDEr/reward histories + configs
experiments/v2_attention_clip_r1/ priority0_report.json	Champion CHAIR + calibration + p(END) audit
experiments/v2_attention_clip_r15/ train_history.json	Per-epoch full-split CHAIR trajectory (degeneration evidence)
experiments/beam_sweep/results.json	17-config beam/ α sweep (Figure 4)
experiments/confidence/temperature.json	Gen-2 NLL-vs-temperature grid + reliability bins
%LOCALAPPDATA%/Temp/ margin_sweep.json	Router margin sweep (Figure 8)
%LOCALAPPDATA%/Temp/ router_margin002_live.json	Live 1,214-image post-fix verification (per-image provider)
backend/inference/ attention_provider.py	Serving code: router, guards, tempering, env knobs
src/components/dashboard/ model-info-view.tsx	Frontend Techniques page mirroring this ledger

Appendix B · Champion hyperparameters (live values)

Parameter	Value	Parameter	Value
Encoder / dim	CLIP ViT-B/16 · 768	Decoder / dim	LSTM · 512
Parameters	8.42M	Vocabulary	BPE-6,000
Embedding dim	256	Max length	38 tokens
CE recipe	bs64 · lr 3e-4 · wd 0.01 · patience 12	GRPO recipe	G=5 · 8 img/step · lr 1e-5 · EMA 0.999
Decoding	beam-5 · α =1.2 · 2-gram log 0.4	Guards	min-len 8 · trim · T=1.3 · margin 0.02
Served checkpoint	v2_attention_clip_r1/best.pt (ep4 EMA)	Seed	42 (all runs)

References

- Bahdanau et al. (2015) — Neural machine translation by jointly learning to align and translate (additive attention).
- Wu et al. (2016) — Google’s NMT: GNMT length penalty (α =0.6–1.2 regime).
- Shao et al. (2024) — DeepSeekMath: GRPO (group-relative policy optimization).
- Radford et al. (2021) — Learning Transferable Visual Models (CLIP), 400M pairs.
- Vijayakumar et al. (2016) — Diverse Beam Search.
- Lin et al. (2014) — Microsoft COCO; Vinyals et al. (2015) — Show and Tell.
- Rossi et al. (2025) — CHAIR: Comprehensive Hallucination Assessment (noun-support metric).
- Guo et al. (2017) — On Calibration of Modern Neural Networks (ECE, temperature scaling).
- Li et al. (2023) — BLIP-2 / Q-Former (referenced in rejected-technique analysis).

Report generated from project artifacts, September 2026 · Muhammad Hamza Mushtaq — Machine Learning Project. All metrics reproducible via the scripts and saved JSONs in Appendix A. Evaluation protocol constant throughout: Flickr8K 1,214-image validation split, beam-5, GNMT α =1.2, seed 42.